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Optimum sizing of a stand-alone hybrid energy system for rural electrification in Bangladesh

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ABSTRACT

This study investigates the potential application of hybridized energy system (i.e., PV/Wind/Diesel) with battery storage in the northern region of Bangladesh. A techno-economic feasibility of different system configurations is evaluated and an optimized system is selected using HOMER (Hybrid Optimization Model for Electric Renewable) software. The hybrid system is optimized to meet a remote stand-alone village community's load demand of 242.56 kWh/day with a 51.52 kW peak load demand. The result of this study indicates that the optimized hybrid system consists of 73 kW PV arrays, a 57 kW Diesel generator, a 387 kWh nominal capacity battery bank, and 28 kW inverters have the minimum Cost of Energy (COE) of 0.37$/kWh and the Net Present Cost (NPC) of $357,284. The outcome also indicates that the optimized system reduces CO_2 emission by around 62% in comparison with the kerosene used in a current situation and 67% with the grid connected system. Furthermore, the Life Cycle Emissions-LCE (kg CO_2-eq/yr) production is considerably lower in PV/Batt/ICE system than the other system configurations. However, this system has a lower Duty Factor (DF) compared to Wind/Batt/Diesel and Batt/Diesel based system. Although this system is not comparable with the grid tariff, the proposed method is economically feasible than solar micro utility system, Wind/Batt/Diesel system, and Diesel generator only system. Additionally, the article discusses the social and economic benefits of implementing the hybridized system along with their barriers and challenges.

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1. Introduction

In the modern era, energy is considering as the measuring scale of the economic growth and the betterment of the individual way of living. The necessity of energy has been rising unprecedentedly all over the world due to the increasing rate of population, the introduction of new technologies, and the growing rate of energy consumption that leads to a massive difference in between the supply chain and demand. Around 75% of the world total energy requirement is satisfied by the fossil fuel (Ahiduzzaman and Islam, 2011). The way of using fossil fuel, the reduction of its sources and its contribution to global warming has made the power generation unsustainable. The ever-increasing problem of Green House Gas (GHG) emissions and at the same time shortage of fossil fuel have promoted the urgency for the sustainable energy technologies to meet the global energy demand (Shezan et al., 2016; Li et al., 2010; Li and Zheng, 2016; Li, 2016; Mandal et al., 2017; Sarker et al., 2018). As a result, energy policies have been revised by taking into account the renewable energy sources to cope with the demand (Ong et al., 2012; Child et al., 2018).

Bangladesh is a highly densely populated country in South Asia with a population of over 160 million (Ghosh and Mandal, 2018). However, only 83% of the country's total people have grid electricity connections which were 38% in the year of 2010 (Farfan and Breyer, 2017; Nandi and Ghosh, 2010a). As per the estimation, the electricity demand is increasing at a rate of 10% per annum and anticipated to be raised in the upcoming years (Hossain and Ahmed, 2013). The electricity demand is expected to be 34,000 MW by the end of the year 2030 (Weaver et al., 2017). Ironically, most of the rural areas are not connected to the national grid, and hence, the Solar Home Systems (SHSs) and the kerosene are the only sources of lighting on an insufficient scale. In this regard, the government of

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Bangladesh is trying to make the electricity connection available for all over the country to meet the increasing energy demand in the domestic sectors. Table 1 represents the power generation statistics in Bangladesh. From the table, it is evident that there is a massive gap between the power demand (11,405 MW) and the electricity served (9,479 MW). Power generation in Bangladesh mostly relies on the natural gas, which is around 64% of total power generation. Moreover, the reserve of natural gas is decreasing at a steady rate with the industrialization and the electrification in the country. However, Bangladesh uses proportionately more fossil fuel, which accounts for around 74% of its total energy consumption than other neighboring countries (Roberts et al., 2018). The transportation and power generation sectors are the two who mainly dependent on the fossil fuel. On the other hand, among the neighboring countries, per capita, electricity consumption in Bangladesh is one of the lowest (311 kWh/capita). Whereas, this figure for India, Pakistan, and Sri Lanka is 806 kWh/capita, 471 kWh/capita, and 535 kWh/capita, respectively (Child et al., 2018). In Bangladesh, rural areas lack electricity supply due to its far distance and dispersing location from the utility grid. In this context, the only way to fulfill their need for electricity is through the diesel generator or the other conventional methods (Lau et al., 2010). As such, the rural families are very dependent on the fuel. Hence, the rising global fuel price has a significant effect on the rural communities. Furthermore, the conventional ways of electricity generation are responsible for the 40% of the total CO_2 emissions (Shrestha et al., 2009). Therefore, only the renewable energy (i.e., solar energy, wind energy, biomass, geothermal, and hydro) is the suitable alternative to reduce the dependency on the fossil fuel to fulfill the electricity demand. In this regard, the Government of Bangladesh introduced Solar Home Systems (SHSs) in off-grid rural areas through the Rural Electrification Board. Halder (2016) reported that around 65,000 SHSs were installed per month and 3,863,964 SHSs were introduced throughout the country. Non-dependency, non-polluting, and availability of lifetime supply make these sources (PV, wind energy) omnipresent and a good contender for alternative energy sources (Bhattacharjee and Acharya, 2015; Zahboune et al., 2016; Ifaei et al., 2017; Al-Falahi et al., 2017; Izadyar et al., 2016). However, the site dependency and the unpredictable nature of the weather are the drawbacks of using these sources as an independent energy provider (Brka et al., 2015). Moreover, the variation of the energy does not match the demand hours which leads to the oversizing of the components causes higher design cost compared to the conventional source (Elhadidy and Shaahid, 2000). Regarding installation cost, the diesel generator is beneficial than the stand-alone solar and wind energy system. Even they cannot perform separately well enough to solve the energy crisis (Bernal-Agustín and Dufo-Lopez, 2009; Maleki, 2017). The issue of the limited fossil fuel, negative consequences of the conventional power sources (Sinha and Chandel, 2015) and the reliability of the separate renewable source to meet the demand lead to a search for the most efficient energy technology such as hybridization of renewable sources with the combustion based prime movers (Adaramola et al., 2014; Tahani et al., 2015; Chowdhury et al., 2015). In this regard, the hybrid energy system is gaining attraction of the researcher's as an efficient alternative to solve the energy crisis in off-grid areas.

Different types of hybrid energy system such as PV/Wind, Wind/Diesel, PV/Wind/Diesel, and PV/Diesel generator along with the storage device battery or without the battery are proven a cost-efficient in some number studies for the remote off-grid areas. To exemplify this, Baneshi and Hadianfard (2016) analyzed the techno-economic feasibility of PV/Wind/Diesel/Batt based hybridized system for a non-residential large electricity application in Southern Iran. They reported that a lower interest rate and carbon tax policy could encourage the sizeable renewable energy based project. Mehrpooya et al. (2018) reported an optimal hybridized system by technical, economic and environmental indicators. However, they did not investigate the Duty Factor (DF, kWh/start-stop/yr), Life Cycle Emissions (LCE, kg CO_2-eq/yr) and sensitivity of cost inputs of different hardware components. Li and Yu (2016) studied a comparative analysis of PV/Batt/Diesel and PV/Batt power system for a household application and found the former method is more economically feasible. Shezan et al. (2016) examined a feasibility analysis of PV/Wind/Batt/Diesel system for the electrification of a remote area. The study revealed that the hybridized system is economically and environmentally feasible compared to the conventional power generation system. They did not consider DF, LCE, sensitivity analysis of input cost components,

Table 1
Power generation statistics of Bangladesh (Child et al., 2018; Farfan and Breyer, 2017; Bangladesh Power Dvelopme, 2017).

Total installed capacity by fuel type (September 2017)		
Fuel type	Installed capacity (MW)	Share of total generation (%)
Natural gas	8709	63.94
Furnace oil	2889	21.21
Diesel	880	6.46
Coal	250	1.84
Hydro electric	230	1.69
Renewables (other than hydro)	3	0.02
Power import	660	4.85
Total installed capacity by plant type (September 2017)		
Plant type	**Installed capacity (MW)**	**Share of total generation (%)**
Combined cycle	4734	34.76
Reciprocating engine	4695	34.47
Steam turbine	2194	16.11
Gas turbine	1105	8.11
Hydro electric	230	1.69
PV	3	0.02
Power import	660	4.85
General statistics		
Installed capacity by public sector, MW (%)		7476 (55%)
Installed capacity by private sector, MW (%)		6145 (45%)
Maximum demand, MW		11405
Maximum demand served, MW		9479
Electricity access to people (%)		83
Electricity consumption, kWh/capita		311

and social perspective in their report. Singh et al. (2017) investigated the PV/FC/Batt system both technically and economically to meet the power demand of an academic research building in Bhopal, India. Rajbongshi et al. (2017) reported that decentralized hybrid energy system (PV/Biomass/Diesel) is an economically viable option for rural electrification where grid extension is not feasible. Moreover, they made a comparison between the grid and off-grid hybrid energy systems for better understanding. Even the hybrid power scheme is more efficient than stand-alone solar PV system which is exemplified in (Abdullah et al., 2010). The result of the study indicates that the effective range of the hybrid energy systems is ∽15%–75% whereas the stand-alone PV system has an efficiency of only ∽10%. Table 2 represents the study of hybrid energy systems with different configurations in the various off-grid area in Bangladesh. It is evident from the Table 2 that the study does not get much attention considering the northern part of the country. Additionally, these studies did not consider DF, LCE, and social perspectives in their analysis. In this context, Nandi and Ghosh (2010b) studied a stand-alone PV/Wind/Diesel generator based hybrid system for the electrification of a remote location, Kutubdia Island, Bangladesh. They found that the Cost of Energy (COE), Net Present Cost (NPC) would reduce at a rate of 20% and diesel usage would cut down to a percentage of 50% than the diesel fuel used in diesel generator based system. However, their study did not include Life Cycle Emissions (LCE), Duty Factor (DF), and social perspective of their system. Another case study illustrated by Nandi and Ghosh (2009) which includes PV/Wind/Batt system for the application of a remote community at Sitakunda, Bangladesh of daily usage of 169 kWh with a peak of 61 kW and found the cost of energy (COE) of 0.363$/kWh. Another contextual analysis was led by Islam and others (Islam et al., 2012) on St. Martin's island in the Bay of Bangladesh. The COE and NPC obtained from the study are 0.345$/kWh and $137,927, respectively with the renewable fraction of 31%. The hybridization of the wind and PV module for meeting up the demand is quite impressive (Nema et al., 2009). An optimization result shows that the PV/Wind/Batt/Diesel generator system is more viable regarding the economy (Kaabeche and Ibtiouen, 2014). Das et al. (2017a) studied a PV/Wind/Biogas/Batt/Diesel generator based hybrid system for stand-alone application in the southern area in Bangladesh. They found that the proposed method has potential benefits of using locally available renewable resources to electrify the remote community than using the diesel generator only and the SHSs. It is evident from the literature review that many researchers have studied the same configurations as considered in this paper. However, the present study investigates the available renewable energy resources in the unelectrified and poverty affected northern area of Bangladesh namely Godagari of Rajshahi Division. Although the DF and the LCE are included in a stand-alone hybridized system using Genetic Algorithm (Das et al., 2017b, 2018; Das and Al-Abdeli, 2017), to the best of author's knowledge based on the available literature, the DF and the LCE have not widely been considered while the hybridized system is optimized using HOMER. Additionally, the effects of installing such projects on the lifestyle of the residents and their economy have briefly discussed. This type of analysis could be beneficial for the policymaker to decide for the electrification of the community like this with the same meteorological conditions.

The primary objectives of this study are (i) to analyze the various system configurations economically and environmentally, (ii) to compare the optimally sized system with the existing national grid and SHSs, (iii) to investigate the consequential performance of different system configurations, and (iv) to study the technical, economic, and social benefits along with the challenges. In achieving this, the present research goes further considering the following factors: Firstly, a feasibility analysis of a PV/Wind/Batt/Diesel based hybrid system of a typical village of Godagari, Rajshahi has been done. Secondly, a comparison between the optimally sized hybrid system with the Solar Home Systems (SHSs) and national grid connection electric supply is carried out both economically and environmentally. Additionally, the system is further analyzed ecologically and compared with kerosene fuel presently used in this area. The consequences of using kerosene and diesel generator for lighting are also discussed to emphasize the importance of this kind of project. Thirdly, the Life Cycle Emissions (LCE, kg-eq CO_2/yr) along with operational emissions (i.e., CO_2, CO, SO_2, NO_x, PM, UHC), Renewable Penetration (RP), Excess Energy (EE), Duty Factor (DF) are examined as well. Fourthly, the research is also extended to analyze the benefits of installing such project on remote community and challenges associated with this by the door to door visiting those studied areas and reviewing the literature on this. This study mainly deals with the available renewable energy resources of this unelectrified areas and their potential application to meet the electrical demand using PV, wind turbine and their hybridization with a diesel generator, and battery storage.

2. Methodology

Conceptual design architecture of the proposed hybrid system articulates in Fig. 1, which consists of PV modules, wind turbine, diesel generator, battery, electric load demand, and converter. On the AC side of the network, diesel generator and load demand are attached, and PV module, wind turbine along with the battery bank is linked with the DC portion. The purpose of the battery bank is to accumulate energy when there is a surplus of power from PV during the daytime and to deliver the stored energy to the system during the load requirement when renewables sources are not producing enough energy, and load demand are below the minimum starting threshold of a diesel generator. In this hybrid system, the DC output from the PV system, wind turbine, and the battery is transformed into AC with the aid of converter.

In this study, an introduction of a well-defined and standardized framework of optimization methodology of a hybrid power system

Table 2
Hybrid energy systems studied in different areas of Bangladesh.

Hybrid energy systems	Storage	Off-grid/grid connected	Study area	Methodology	Performance indicators
PV/Batt/Diesel (Nandi and Ghosh, 2010a)	Lead acid	Off-grid	East-southern	HOMER	NPC, COE, GHG emissions, sensitivity analysis
PV/Batt/Diesel (Bala and Siddique, 2009)	Lead acid	Off-grid	East-southern	GA	NPC, CO_2 emissions
PV/Wind/Batt (Nandi and Ghosh, 2009)	Lead acid	Off-grid	East-southern	HOMER	NPC, COE, CO_2 emissions
PV/Batt/Biogas (Rahman et al., 2014)	Lead acid	Off-grid	Central	HOMER	COE, TAC, sensitivity analysis
PV/Batt/Biogas (Khan and Martin, 2015)	Lead acid	Off-grid	South-west	HOMER	COE, Payback period, sensitivity analysis
PV/Tidal/Batt/Diesel (Das et al., 2016)	Lead acid	Off-grid	Southern	HOMER	NPC, COE, emissions
PV/Wind/Batt/Diesel (Rashid et al., 2017)	Lead acid	Off-grid	Southern	HOMER	NPC, COE, emissions
PV/Wind/Batt/Biogas/Diesel (Das et al., 2017a)	Lead acid	Off-grid	Southwestern	HOMER	NPC, COE, Payback period, emissions
PV/Biomass/Batt/Diesel (Islam et al., 2018)	Lead acid	Off-grid	North-east	HOMER	NPC, COE, Payback period, emissions

Table 3
Estimated load demand of the selected village.

Load	Power Rating (W)	Summer (March to October)						Winter (November to February)					
		Low-income household (110)		Medium income household (60)		High-income household (20)		Low-income household (110)		Medium income household (60)		High-income household (20)	
		No. of items	h/day	No. of items	h/day	No. of items	h/day	No. of items	h/day	No. of items	h/day	No. of items	h/day
Energy efficient bulb	20	2	5	3	5	4	6	2	5	3	5	4	6
Fan	70	1	5	2	6	2	8	–	–	–	–	–	–
TV	80	–	–	1	6	1	6	–	–	1	6	1	6
Fridge	150	–	–	–	–	1	24	–	–	–	–	1	24
Mobile Charger	25	1	3	2	3	3	3	1	3	2	3	3	3
Total Demand		**293 (kWh/day)**						**182 (kWh/day)**					

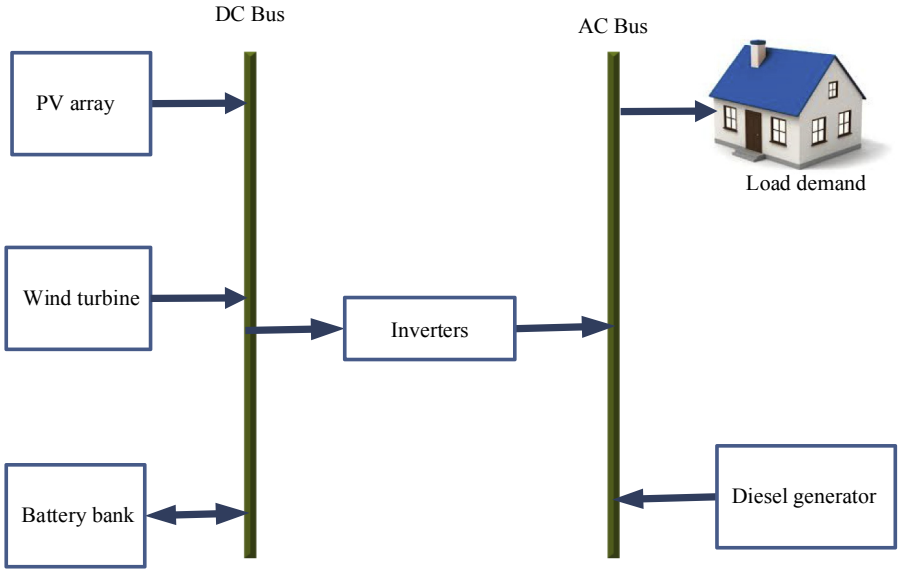


Fig. 1. Block diagram of the proposed hybrid system.

is utilized to meet the load requirements of the off-grid region as shown in Fig. 2. HOMER works in three different phases. Firstly, before performing HOMER analysis, the assessment of the layout of system configuration (Fig. 1), electric load requirements, and available renewable energy sources (solar irradiation and wind data) and temperature data of the selected site is carried out. The technical and economic details of hardware components as presented in Table 4 are inserted into HOMER software. Secondly, with the given above data, HOMER software performs the analysis and provides a set of system configurations based on the lowest Cost of Energy (COE, $/kWh), and NPC ($). In this regard, details modeling of technical and economic data of each configuration are necessary to satisfy the user's specified load requirements and other technical constraints such as fuel cost, excess energy. Thirdly, in the post HOMER analysis, the outcomes of different systems are compared, and the optimized system is chosen based on technical and environmental preferences. The essential performances regarding LCE, RP, DF, and social perspectives are also examined. Although HOMER does not give any information for social issues, this study discussed the perceptions based on the optimized cost and environmental data while comparing the existing scenarios.

2.1. Demographic profile and meteorological data of the studied area

The village area which is taken under consideration in this study is situated in Godagari of Rajshahi division lacking the service of electrical connection. The geographical position and the aloofness from the central city outplayed the possibility of being furnished with the proper electrification system. The village, Chorasariadho, is situated at 8820.2'E and 2427.5'N. There are 220 households with an average of six persons per household in the village. As in the village supply of electricity is not available, kerosene and candles are the only sources for illuminating and biomass such as forest residue, agricultural residues, and cow dung are used for cooking purpose. However, the village has the different resources of potential renewable energy. Most of the villagers are fisherman, and daily workers and others live on small business. The value of annual mean solar irradiation is 4.84 kWh/m^2/day, and the clearness index is 0.65 for the selected area. Fig. 3 represents the monthly solar radiation of the selected area. The monthly average wind resource data is selected directly using the National Aeronautics and Space Administration (NASA) Surface meteorology and Solar Energy database (Bhattacharyya, 2012). The mean annual wind velocity is 2.90 m/s. According to the HOMER software, July gives the maximum wind velocity which is 3.95 m/s and December stands for the minimum wind velocity of 2.23 m/s as shown in Fig. 4 that exemplifies the Monthly average wind velocity of the selected village. The output power from solar irradiation is dependent on the atmospheric temperature. April or May gives the maximum reading of solar radiation and atmospheric temperature. Fig. 5 is for the average monthly ambient temperature for the selected area. Power generation from wind turbine varies along with the wind velocity as it does not remain constant all-round the year.

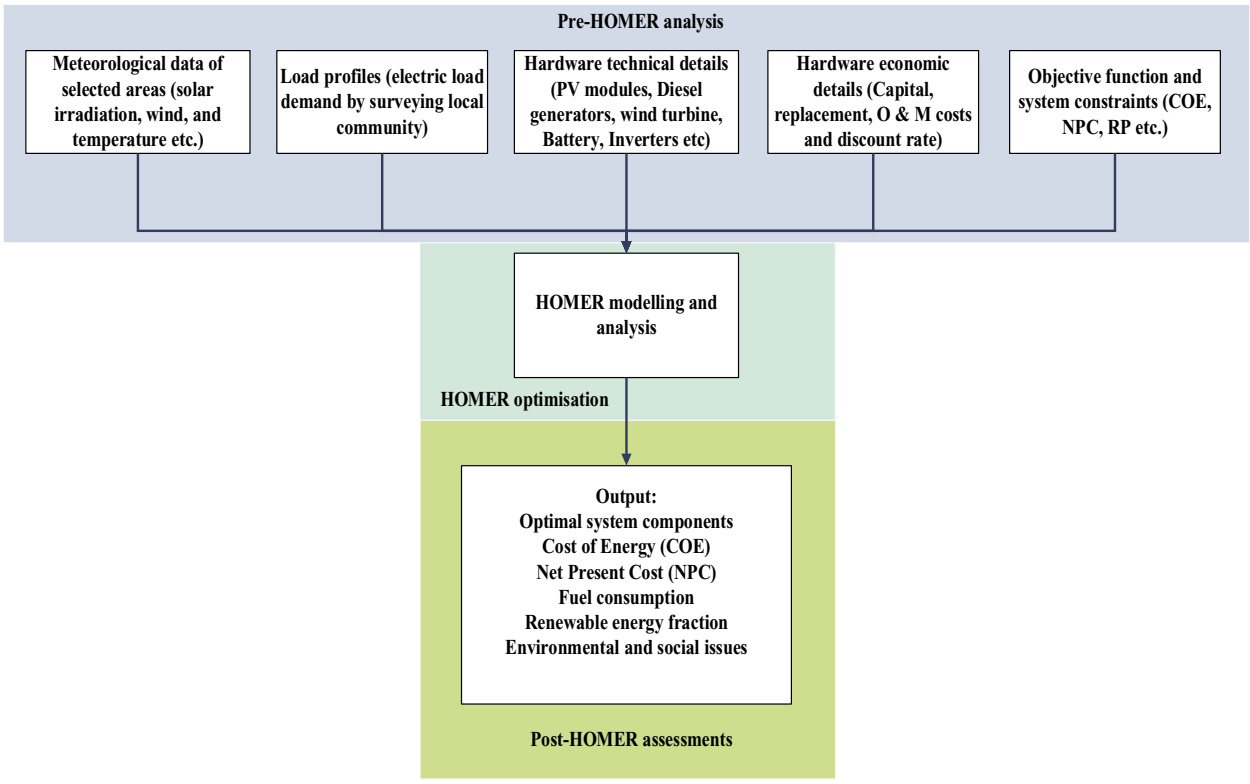


Fig. 2. Working framework for Optimization using HOMER.

Table 4
Technical and economic data for hardware components used as an input for this simulations (Brka et al., 2015; Rajbongshi et al., 2017; Das et al., 2017a; Das and Al-Abdeli, 2017; Maleki and Askarzadeh, 2014).

Components	Technical description	Capital cost ($)	Replacement cost ($)	O & M cost ($)	Lifetime (yr)	LCE (kg CO_2-eq/kWh)
PV Module	327 W	1300/kW	0	20/yr	25	0.045
Diesel Generator	57 kW, 50 Hz	370/kW	296/kW	0.05/h	15,000 h	0.88
Wind turbine	1 kW, 12–48 VDC Nominal.	4450	4450	110	20	0.011
Lead-acid battery	6 V, 820 Ah (4.92 kWh)	1100	1000	10/yr	12	0.028
Inverter	1 kW	800	750	8	15	0
Discount rate (%)	10					
Fuel cost ($/l)	0.91					

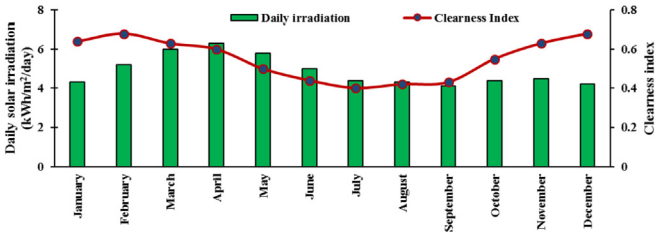


Fig. 3. Monthly average solar irradiation of selected area.

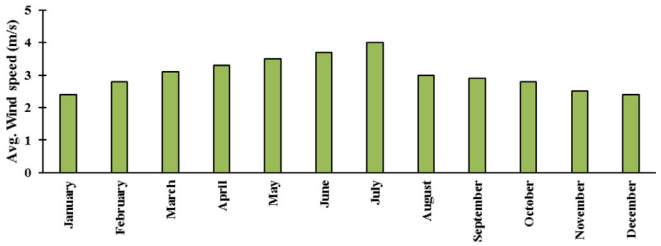


Fig. 4. Monthly average wind velocity of the selected area.

2.2. Estimation of electrical load of the studied area

In this project, mainly light, fan, mobile charger, television, and fridge are considered as electrical load necessitated for domestic purposes (Table 3). The total electrical load demand for the residential area is determined by the data measurement on a daily hourly basis and also by surveying electric supply and distribution authority, village chairman, school teachers, local people, and workers. It has also been consulted previously published literature about rural Bangladesh (Nandi and Ghosh, 2009) and triangulated with expert opinions and personal judgments. The load demand is mainly estimated based on the two season winter (November to February) and summer (March to October). Patterns of the household works and the commercial activities are the main criteria for selecting the load. Therefore, for the winter season, in fact, there will be no loads of fan and energy will consume by bulb, TV, mobile charger and fridge (Table 3). The total projected energy demand of the area for the winter season is 182 kWh/day whereas for the

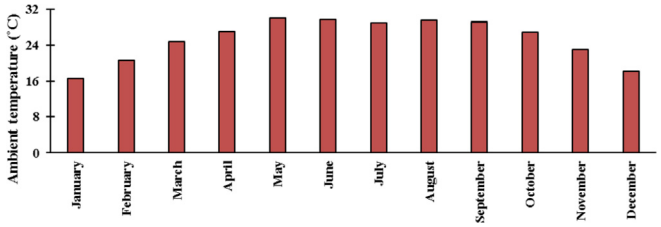


Fig. 5. The average monthly ambient temperature of the selected area.

summer season it increases to 293 kWh/day (Table 3). Fig. 6 represents the daily load profile for the summer (March–October) and winter (November–February) season. During the summer season, fan, light, and TV consume most of the electric energy. Therefore, the demand for electricity for this season is comparatively higher than other seasons.

2.3. Modeling of solar PV for hybrid energy system

The performance of a solar PV module varies due to the different operating conditions and environmental conditions. In this study, Sun Power made commercially available monocrystalline PV module used (Model: E20-327) (Askarzadeh, 2017). The PV power output varies according to the solar irradiation incident on it and the environmental factors (i.e., ambient temperature, wind velocity) (Radziemska and Klugmann, 2002). It is evident that annual output energy by a PV is possible to increase with the 0.1%/°C increase of the temperature coefficient. In this context, the PV power output on an hourly basis is calculated by using the Equation (1) (Khalid et al., 2017), whereby Y_{PV} (kW) is the rated capacity of PV array, f_{PV} (%) is the derating factor, I_T (kW/m^2) is the solar irradiation incident on the PV array, I_S (kW/m^2) is the incident solar irradiation at standard test conditions, α_P is the temperature coefficient of power, T_c (℃) is the PV cell temperature, and T_s (℃) is the PV cell temperature under standard test conditions (25 °C).

$$P_{PV} = Y_{PV} f_{PV} \left(\frac{I_T}{I_S}\right)[1 + \alpha_P(T_c - T_s)] \quad (1)$$

The cell temperature T_c (℃) can be obtained from the energy balance of the PV module using Equation (2), where $(\tau\alpha)$ is the effective transmittance-absorptance of PV array, η_{PV} (%) is the PV panel efficiency, U_L (kW/m^2 ℃) is the heat transfer coefficient, T_a (℃) is the ambient temperature.

$$(\tau\alpha)I_T = \eta_{PV} I_T + U_L(T_c - T_a) \quad (2)$$

Equation (2) rewrites as follows:

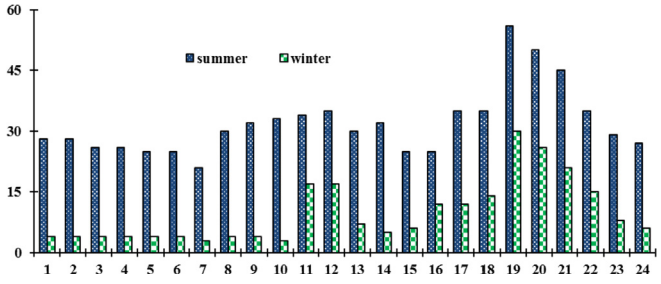


Fig. 6. Daily load profile for the summer (March–October) and winter (November–February) season.

$$T_c = T_a + I_T\left(\frac{\tau\alpha}{U_L}\right)\left(1 - \frac{\eta_{PV}}{\tau\alpha}\right) \quad (3)$$

However, the value of $\left(\frac{\tau\alpha}{U_L}\right)$ is challenging to measure and therefore, based on the manufacturer report on the Nominal Operating Cell Temperature (NOCT) which results at 800 W/m^2 solar irradiation, at 20 °C ambient temperature, and at no load condition (i.e., $\eta_{PV} = 0$). The Equation (3) can be re-written as follows:

$$\left(\frac{\tau\alpha}{U_L}\right) = \frac{T_{c,NOCT} - T_{a,NOCT}}{I_{T,NOCT}} \quad (4)$$

Therefore, it is possible to obtain the final PV cell temperature by using the following Equation, whereby HOMER assumes the value of $(\tau\alpha)$ as 0.9.

$$T_c = T_a + I_T \frac{T_{c,NOCT} - T_{a,NOCT}}{I_{T,NOCT}}\left(1 - \frac{\eta_{PV}}{0.9}\right) \quad (5)$$

In this system, no tracking arrangements have been considered and 80% de-rating factor for each panel is taking into account due to changing consequences of temperature and debris on the solar panels.

2.4. Modeling of wind turbine for hybrid energy system

HOMER gives the total capacity of the wind turbine based on the minimum NPC and COE. In this study, depending on the wind velocity characteristic, BWC EXCEL 1-R (1 kW capacity) wind turbine system is considered with 48 V DC power system, which is connected to the DC bus. The wind turbines are selected based on the service time, hub height and cost of the component, type of electricity generated, and cut-in wind speed. As such, power output performance is different from different wind generators and depends on the wind speed at hub height and the speed characteristics of the turbine. Fig. 7 illustrates the wind turbine characteristics curve where V_1 (m/s) refers to cut-in speed, V_r (m/s) is the rated speed, and V_2 (m/s) is the cut-out speed. The required wind speed at a specific site at certain hub height is determined by using the Power law Equation (Deshmukh and Deshmukh, 2008), whereby V (m/s) and V_{ref} (m/s) are the wind speed at the hub and reference height H (m) and H_{ref} (m), respectively. Additionally, γ is the ground surface friction coefficient and its value varies from less than 0.10 for very flat land, water to more than 0.25 for heavily forested landscapes (Kaabeche and Ibtiouen, 2014; Al-Sharafi et al., 2017).

$$V = V_{ref}\left[\frac{H}{H_{ref}}\right]^{\gamma} \quad (6)$$

The wind turbine power output is expressed as follows (Deshmukh and Deshmukh, 2008),

$$P_w(V) = \begin{vmatrix} 0 & \text{for,} & V < V_1 \\ aV^3 - bP_r & \text{for,} & V_1 < V < V_r \\ P_r & \text{for,} & V_1 < V < V_2 \\ 0 & \text{for,} & V > V_2 \end{vmatrix} \quad (7)$$

where V_1 (m/s) is the cut-in speed, V_2 (m/s) is the cut-off speed and V_r (m/s) is the rated speed of the wind turbine. The rated power is P_r (kW). The value of constant a and b is calculated using Equations (8) and (9) respectively.

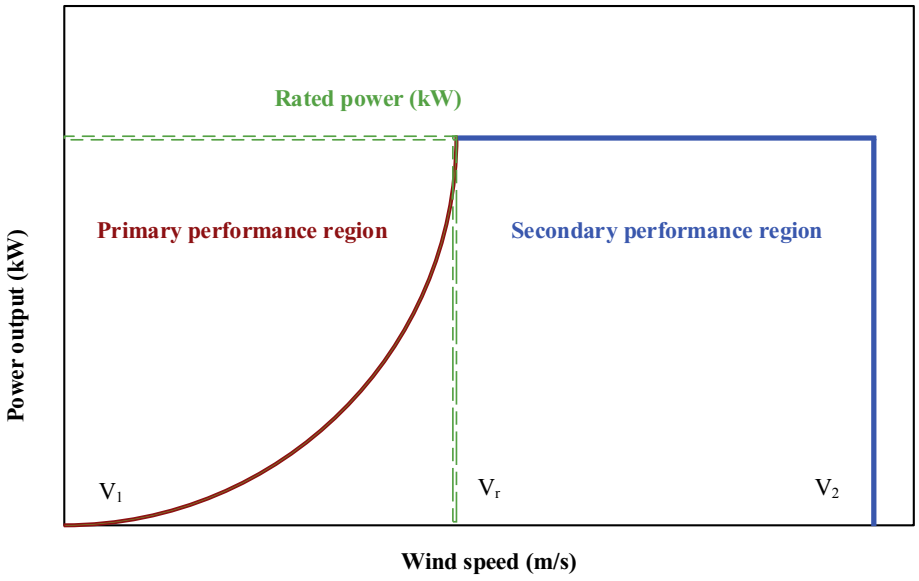


Fig. 7. Power characteristics curve of the wind turbine.

$$a = \frac{P_r}{\left(V_r^3 - V_1^3\right)} \tag{8}$$

$$b = \frac{V_1^3}{\left(V_r^3 - V_1^3\right)} \tag{9}$$

Generally, the value of V_1 varies from 2.5 to 3.5 m/s and the value of V_2 varies from 20 to 25 m/s (Kaabeche and Ibtiouen, 2014). However, the actual electric power output from the wind turbine is calculated using Equation (10) (Deshmukh and Deshmukh, 2008), where A_w (m^2) is the swept area, η_w (%) indicates the wind turbine generator and corresponding converter efficiency.

$$P_{wt} = P_w A_w \eta_w \tag{10}$$

2.5. Modeling of diesel generator for hybrid energy system

In this study, the purpose of the diesel generator is to supply the necessary power demand when the renewables (i.e., PV and wind) and battery bank are not able to meet the load demand. HOMER calculates the capacity of the diesel generator set based on the peak electric demand (peak demand plus 10%) as the auto-size generator is selected. In this context, a (peak demand $52 \times 1.1 = 57$) 57 kW diesel generator is suitable for this hybrid system along with the lifetime of 15000 h. The efficiency of a diesel generator is considered as 35%. The maximum load ratio of the generator is 25%. The associated with the fuel consumption for the electricity production is calculated as follows (Khalid et al., 2017),

$$F = F_{0,dg} Y_{dg} + F_{1,dg} P_{dg} \tag{11}$$

where F_0 is the fuel curve intercept coefficient (0.000205m^3/h), F_1 is the fuel curve slope (0.00025 m^3/h/kW), Y_{dg} (kW) is the rated capacity of the generator and P_{dg} (kW) is the electric output of the generator. The lower heating value of diesel fuel is 43.2 MJ/kg with a density of 820 kg/m^3 (Khalid et al., 2017).

In the study which follows, the Duty Factor (DF) (kWh/start-stop/yr.) is the ratio of energy generation from supplementary prime movers to the total start-stop and can be calculated by using Equation (12), where $N_{s/s}$ is the number of start-stop (Das et al., 2017b).

$$DF = \frac{P_{dg}}{N_{s/s}} \tag{12}$$

2.6. Modeling of battery for hybrid energy system

The purpose of the battery bank is to store extra power produced by the PV and wind turbine and to deliver the power when the PV and wind turbine fails to meet up the required load demand. Diesel generator comes into action when all three fails to fulfill the necessary load demand. For the longevity of the battery bank, the maintenance of the battery charge within 20% is very necessary (Das et al., 2017b). In this study, **Surrette 6CS25P** is chosen as a storage battery, which has the nominal voltage of 6 V, the theoretical capacity of 6.91 kWh, the maximum capacity of 820 Ah and capacity ratio of 0.237. The round trip efficiency of the battery bank is 80% (Li et al., 2013).

Here, with the help of Equation (13) the value of battery energy can be calculated (Brka et al., 2016), where $Q_{bat,0}$ (kWh) is the initial battery charge, V_{bat} (V) is the battery voltage and V_{bat} (A) is the battery current.

$$Q_{bat} = Q_{bat,0} + \int_0^{\tau} V_{bat} I_{bat} dt \tag{13}$$

The state of battery charge is defined by the Equation (14)

$$B_{soc} = \frac{Q_{bat}}{Q_{bat,\max}} \times 100(\%) \tag{14}$$

The maximum allowable battery charge power and discharge power of battery bank is calculated using battery kinetic model. The maximum amount of power that can be absorbed by the two-tank system is given by the following Equation.

$$P_{batt,\mathrm{cmax},kbm} = \frac{kQ_1e^{-k\Delta t} + Qkc\left(1 - e^{-k\Delta t}\right)}{1 - e^{-k\Delta t} + c\left(k\Delta t - 1 + e^{-k\Delta t}\right)} \quad (15)$$

where, Q_1 (kWh) is the available energy in the battery at the beginning of the time step, Q (kWh) is the total amount of power in the battery at the beginning of the time step, k is the battery rate constant (2.12 h^{-1}), c is the battery capacity ratio (0.305), and Δt (h) is the length of the time step. On the other hand, the maximum amount of power that the battery bank can discharge over a specific length of time is given by the Equation (16), whereby Q_{max} is the total capacity of the battery bank,

$$P_{batt,d\mathrm{max},kbm} = \frac{-kcQ_{\max} + kQ_1e^{-k\Delta t} + Qkc\left(1 - e^{-k\Delta t}\right)}{1 - e^{-k\Delta t} + c\left(k\Delta t - 1 + e^{-k\Delta t}\right)} \quad (16)$$

HOMER uses the Kinetic Battery Model (Manwell and McGowan, 1993) to determine the amount of energy that can be absorbed by or withdrawn from the battery bank each time step. The maximum charge power and discharge power gives the allowable range for the power into and out of the battery bank in any time step. Once HOMER calculates the actual charge or discharge power, it calculates the resulting amount of available and bound energy at the end of the time step using the Equations (17) and (18) (Manwell and McGowan, 1993).

$$Q_{1,end} = Q_1e^{-k\Delta t} + \frac{(Qkc - P)\left(1 - e^{-k\Delta t}\right)}{K} + \frac{Pc\left(k\Delta t - 1 + e^{-k\Delta t}\right)}{K} \quad (17)$$

$$Q_{2,end} = Q_2e^{-k\Delta t} + Q(1 - C)\left(1 - e^{-k\Delta t}\right) + \frac{P(1 - c)\left(k\Delta t - 1 + e^{-k\Delta t}\right)}{K} \quad (18)$$

where the available bound energy at the end of the time step is $Q_{1,end}$ (kWh), $Q_{2,end}$ (kWh), and P is the power into (positive) or out of (negative) the battery bank. More details are available in the literature (Manwell and McGowan, 1993), and in the HOMER Pro, microgrid analysis tool Help section.

2.7. Modeling of inverter for hybrid energy system

The main purpose of the inverter is to change into the AC current for AC loads from the DC current DC bus. The following Equation (Brka et al., 2015) is using to determine the load side power, whereby, P_{in} (kW) is the DC power input to the inverter, P_{out} (kW) is the AC output power and η_{inv} (%) is the inverter efficiency (95%).

$$P_{in} = \frac{P_{out}}{\eta_{inv}} \quad (19)$$

2.8. Economic criteria for hybrid energy system

HOMER simulation is not possible without the financial parameters. This study consists of annual real interest rate, Net Present Cost (NPC), Cost of Energy (COE), initial capital cost, and replacement cost for the different system configurations.

2.8.1. Net present cost (NPC)

The total net present cost is the combination of initial capital cost, replacement cost, operation and maintenance costs and energy costs (fuel cost plus any associated costs) over its entire lifetime in this project. The HOMER simulation results are organized based on the NPC which is found by using the Equations as follows (Das et al., 2017a) (Ngan and Tan, 2012),

$$C_{NPC} = \frac{C_A}{CRF(i,N)} \quad (20)$$

$$CRF(i,N) = \frac{i(1+i)^N}{(1+i)^N - 1} \quad (21)$$

$$i = \frac{i' - f}{1 + f} \quad (22)$$

In the above Equations, C_A is the total annualized cost ($/year), CRF is the capital recovery factor, i (%) is the annual real interest rate, i' (%) the nominal interest rate, f (%) is the annual inflation rate and N is the project lifetime (in the year). In this study, a discount rate of 8% and an annual inflation rate of 2% are considered.

2.8.2. Cost of energy (COE)

The Cost of Energy (COE) is an essential economic criterion for optimal sizing for a hybridized power system which by definition is the mean cost per kilowatt-hour ($/kWh) of useful electrical energy. It is the ratio of the system components annualized cost to the full energy generation, which supports the demand (excludes excess heat). The Equations (23) and (24) represents the calculation of COE (Hiendro et al., 2013),

$$COE = \frac{C_A}{E_S} \quad (23)$$

$$C_A = C_{A_cap} + C_{A_rep} + C_{A_O\&M} \quad (24)$$

where, C_A ($) is the total annualized cost including annualized capital cost (C_{A_cap}), annualized replacement cost (C_{A_rep}), annualized O&M ($C_{A_O\&M}$) cost, and E_S is the energy served in a year.

2.8.3. Life Cycle Emissions (LCE)

By definition, the LCE (kg CO_2-eq/yr) is the equivalent CO_2 emissions from the energy used to manufacture, transport, and recycle the components of system configurations as well as the operational emissions. The following Equation presents the mathematical expression for the calculation of LCE (Brka et al., 2015):

$$LCE = \sum_{i=1}^{x} \beta_i E_L \quad (25)$$

where x is the number of components, β_i (kg CO_2-eq/kWh) is the lifetime equivalent CO_2 emissions of each component, E_L (kWh) is the energy generation or stored in each component. The lifetime equivalent CO2 emissions (kgCO_2-eq/kWh) of different components used in this simulation are reported in Table 4.

In this study, the design of the hybrid system is done by entering the required resource parameters into the HOMER software. Table 4 represents the technical and economic data of different components utilized as an input data for the simulations.

3. Results and discussions

In the simulation process, HOMER estimates the cost and at the same time determine the feasibility of hybridized systems over the year with a list of system configurations and their capacities are sorted based on lowest COE and NPC. In this study, among many

Table 5
Simulation results summary of different categorized system configurations.

Characteristics	Case I	Case II	Case III	Case IV	Case V	Case VI
Diesel generator capacity (kW)	57	57	–	–	57	57
PV Capacity (kW)	73	73	154	225	–	–
Wind turbine capacity (kW)	–	1	–	1	–	1
Inverter capacity (kW)	28	29	52	54	13	13
Number of battery	56	54	159	106	31	31
Capital costs ($)	200,256	202,280	415,957	455,845	65,359	69,747
Replacement costs ($)	56,780	65,913	102,732	74,047	84,577	85,429
Operating costs ($)	51,941	52,386	54,913	64,769	132,760	133,003
Fuel costs ($)	4717	4831	–	–	30,716	30,692
Excess energy (kWh/yr)	25,835	25,796	137,048	249,478	412	419
Unmet load (kWh/yr)	0	0	62.7	64.4	0	0
Fuel consumption (m^3)	5.24	5.37	–	–	34.13	34.10
Diesel engine operating hour	896	917	–	–	4161	4167
Engine starts/yr	435	445	–	–	828	830
Duty Factor (kWh/start-stop/yr)	31	31			123	123

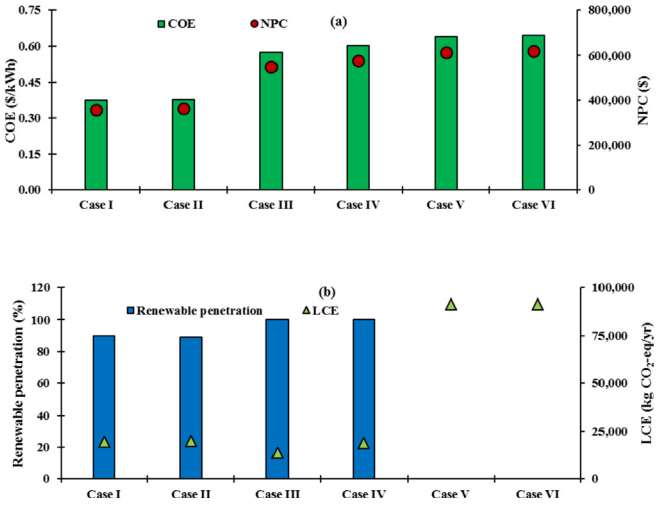


Fig. 8. Comparison between (a) Cost of Energy (COE), Net Present Cost (NPC), (b) Renewable Penetration (RP), and Life Cycle Emissions (LCE) for different hybridized system configurations.

configured energy systems, six different scenarios are evaluated to find the optimized system configuration both technically and economically as presented in Table 5. The proposed hybrid energy systems have been mainly focusing on Cost of Energy (COE), Net Present Cost (NPC), Renewable Penetration (RP), Excess Energy (EE), Life Cycle Emissions (LCE) and operational emissions, and Duty Factor (DF). Although HOMER simulated various components of the energy system, however, it only exhibits the executable hybrid systems scenarios. In the following sub-sections, different situations are discussed and compared based on the costs, environmental emissions, and the consequential performances to find the optimal system configuration.

3.1. The case I: PV/Batt/diesel generator hybrid system

In this study, the first configuration is considered as PV/Batt/Diesel generator based hybrid system. From Fig. 8(a), it is clear that this system has the lowest COE of 0.37$/kWh and the NPC of $357,284 which consists of a 57 kW diesel generator, 73 kW PV module, 56 batteries, and 28 kW inverters. Although the capital cost is the fourth highest, the replacement cost and the operating cost is the lowest among the six categorized configurations as shown in Table 5. The annualized cost is lower than other scenarios (Table 6). Results from Table 8 indicate that the emissions of CO_2 (13,720 kg/yr) are quite low than the national grid (56,662 kg/yr), Diesel only (89,338 kg/yr), and the kerosene-based (36,135 kg/yr) lighting systems because of the higher penetration of the renewable energy which is 90% of the total power delivered. The total power generated from this scenario is 128,602 kWh. The contribution from the PV is 89% of the total power and the rest 11% comes from the diesel generator (Fig. 9). It is also evident from the result that the excess energy production is 20% of the total energy generation and this system is used to fully satisfied with the load requirement. Additionally, the duty factor for the diesel engine is 31 kWh/start-stop, and the LCE is second lowest among the system configurations. In summary, the PV/Batt/Diesel has lowest COE and NPC among the system configurations and fully capable of meeting the load requirements of the studied area. Additionally, this system releases less CO_2, LCE, and other emissions to the atmosphere in comparison with the other cases (Table 7).

3.2. Case II: PV/wind/Batt/diesel generator hybrid system

In this scenario, the hybrid system consists of 73 kW PV module, 54 batteries, a 1 kW wind turbine, and a 57 kW diesel generator (Table 5). This system is configured with a small capacity of the wind turbine (1 kW) due to the availability of wind velocity (average 2.9 m/s only) of the selected area and the higher capital cost. In this context, HOMER selects only one wind turbine along

Table 6
Total annualize cost of different hybridized systems.

System configurations	Capital cost ($)	Operating cost ($)	Replacement cost ($)	Salvage ($)	Resource ($)	Total cost ($)
Case I	18,509	4801	5248	−252	4717	33,023
Case II	18,696	4842	6092	−978	4832	33,484
Case III	38,446	5075	9495	−2221	0.00	50,795
Case IV	42,132	5986	6844	−1594	0.00	53,369
Case V	6041	12,271	7817	−105	30,717	56,740
Case VI	6446	12,293	7896	−157	30,692	57,170

Table 7
Operational emissions from different system configurations.

Operational emissions (kg/yr)	Case I	Case II	Case III	Case IV	Case V	Case VI
CO_2	13,720	14,052	–	–	89,338	89,266
CO	86.5	88.6	–	–	563	563
SO_2	33.6	34.4	–	–	219	219
NO_x	81.2	83.2	–	–	529	529
UHC	3.77	3.87	–	–	24.6	24.6
PM	0.524	0.537	–	–	3.41	3.41

Table 8
Operational CO_2 emissions comparison between different systems.

Scenarios	CO_2 emissions (kg/yr)
PV/Batt/Diesel (Optimal system)	13,720
National grid	41,085
Diesel only	89,338
Kerosene	36,135

with the PV, Diesel generator and battery storage to meet the load demand with the possible minimum cost. Although this system is comparable to Case I regarding COE, and NPC, this system has slightly higher capital cost and 16% higher replacement cost. The contribution of energy from PV modules and diesel generator to the total energy generation are around 89% and 11%, respectively. However, the energy generation by the wind turbine to the total energy is <1% as shown in Fig. 9. The rate of operational pollutant emissions (except NO_x) from this configuration is higher than the Case I. This system, however, has comparable EE, DF, and LCE with the Case I. From the above discussion, it is evident that the Case II is quite similar to Case I both economically and environmentally. Ironically, marginally higher capital and replacement cost and a small contribution of wind power, makes Case II complicated and not more attractive as Case I.

3.3. Case III: PV/Batt hybrid system

The third economically feasible system is the PV/Batt hybrid system consists of 154 kW PV module and 159 batteries which is the most significant number of batteries required by any hybrid system in this study. This system has COE of 0.57$/kWh. It is also evident from the Table 5 that the capital cost ($415,957) and the replacement cost ($102,732) is near about twice than the previous two cases. The total NPC ($549,572) is almost 50% higher than the last two configurations whereas this configuration contributes zero operational emissions with the highest renewable penetration of 100% (Fig. 8(b)). Although the annualized operating cost is comparable, this system has higher capital and replacement cost (from the battery) as compared to Case I and Case II. However, the LCE for this system is 13,851 kg-eq CO_2/yr. The PV module generates the total energy, which is 241,796 kWh as shown in Fig. 9. The results also reveal that the PV/Batt hybrid system generates higher excess energy compared to previous two cases. This system has zero operational emissions and lowest LCE with the highest renewable contribution among the Cases with the cost of the COE and the NPC than the Case I and Case II. Additionally, a large number of batteries requires frequent maintenance which makes the system infeasible for a remote area like this.

3.4. Case IV: PV/wind/Batt hybrid system

The research has extended to analyze the system with PV/Wind/Batt hybrid which is in the fourth feasible configuration regarding COE at 0.60$/kWh and the NPC at $577,418. It includes the 225 kW PV module, 1 kW wind turbine, and 106 batteries with the capital cost of $455,845, operating cost of $64,769, and the replacement cost of $74,047. The highest capital cost makes the configuration economically infeasible. From Fig. 8(a), it is evident that the COE and NPC are quite comparable with the Case III but way higher than the first two cases. Although the COE and renewable penetration (100%) is comparable, the LCE (18,786 kg-eq CO_2/yr) is higher than the Case III (13,851 kg-eq CO_2/yr). It is evident from the outcome that, the fully renewably powered system (Case III and Case IV) is not economically feasible because of the higher cost of energy compared to the system supplemented by the diesel engine (Case I and Case II).

3.5. Case V: Batt/diesel generator system

The study also investigates the Batt/Diesel generator system including a 57 kW diesel generator and 31 batteries, which is the lowest amount of cell utilized by a system. The COE and NPC are

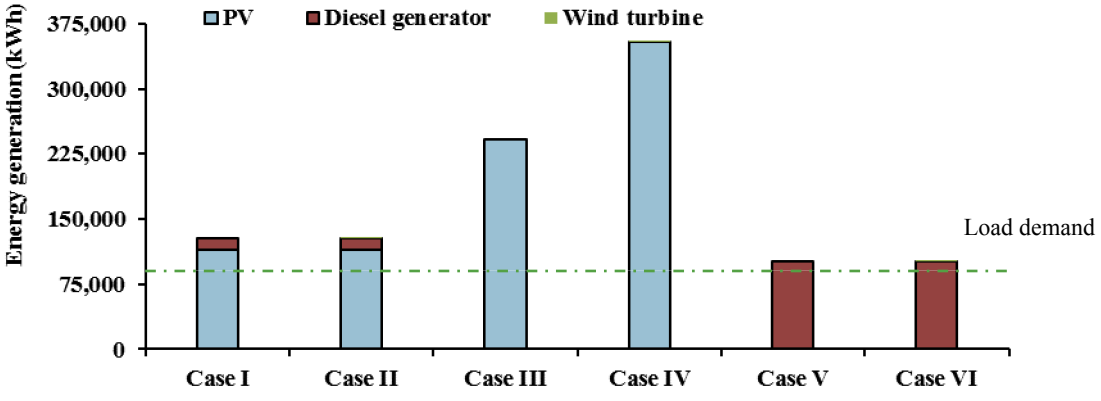


Fig. 9. The share of energy generation from different components for different hybridized system configurations.

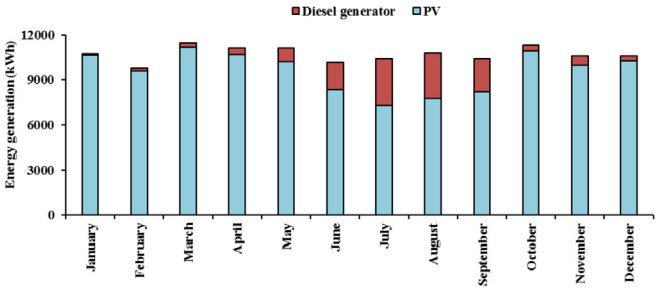


Fig. 10. The share of monthly energy generation from different components of the optimized hybrid system (PV/Batt/Diesel generator).

higher than the PV/Batt/Diesel and the PV/Wind/Batt/Diesel systems but comparable to PV/Batt and PV/Wind/Batt systems as shown in Fig. 8(a). Although the system has lower capital cost, the replacement cost and the operating cost are much higher than the above system configurations. Meanwhile, this system mainly depends on the diesel generator. Therefore the fuel cost ($30,716) is found to be the highest. The total energy generated by the system is quite low than the other cases where the diesel generator shares the all (Fig. 9). The configuration is unfavorable because of the highest operational emissions (Table 7) as well as the Life Cycle Emissions (Fig. 8 **(b)**) with 0% renewable penetration. This system has the highest duty factor among the system configurations as the power generation solely relies on the diesel generator.

3.6. Case VI: wind/Batt/diesel generator hybrid system

The last analysis is based on the Wind/Batt/Diesel hybrid system, which is the combination of a 1 kW wind turbine, a 57 kW diesel generator, and 31 batteries with the highest operating cost of $133,003, the replacement cost of $85,429, and fuel cost of $30,692 (Table 5). This system is not economic because of the higher annualize cost (Table 6) and is not the environmental friendly either due to the extreme rate of pollutant emission from the hybrid system (Table 7 and Fig. 8(b)). Moreover, the highest COE and the NPC are the reason why this configuration is entirely infeasible (Fig. 8(a)). From the Fig. 9, it is evident that the diesel generator is the only source of energy generation. As discussed above, the wind turbine contributes a small portion of total energy generation.

3.7. Optimal configuration (case I: PV/Batt/diesel)

In this study, economic indicators and environmental emissions are the two criteria based on which optimal configuration has been selected. Among the six cases, the configuration of Case I is considered as the most cost-effective hybrid system because of its low Cost of Energy (COE) of 0.37$/kWh and NPC of $357,284. Although, Case II gives the appropriate results in terms of COE, the replacement and operating costs, and emissions are higher than the Case I. From the Table 6, it is apparent that, Case I has the slightly higher annualized capital cost than the Case V and Case VI but has the lower operating and replacement cost among all the six system configurations. It is also evident that the operational CO_2 emissions (13,720 kg/yr) and the Life Cycle Emissions (19,668 kg-eq CO_2/yr) of Case I are significantly lower than Case V (89,338 kg/yr and 91,563 kg-eq CO_2/yr, respectively) and Case VI (89,266 kg/yr and 91,414 kg-eq CO_2/yr, respectively). Fig. 10 is the representation of monthly energy generation by the different components (i.e., PV and Diesel generator) of the optimized hybrid energy system (PV/Batt/Diesel generator). The contribution of PV is maximum in every month energy generation whereas the contribution of a diesel generator is increasing slowly from January up to July then it is again started decreasing. Additionally, Fig. 11 is the representation of different hardware components (i.e., PV, Diesel generator, and battery) which are working together to satisfy the load demand over a week of July. It is evident from the Fig. 11 that maximum share of load demand is fulfilled by the PV and in absence of PV, battery bank is used to satisfy the load requirements. However, when neither solar PV nor battery bank is able to meet the demand, the diesel generator is introduced to meet the demand. Fig. 12 **(a)** shows the State of Charge (SOC) of the battery over the year, whereas Fig. 12 (b) depicts the SOC over a month for PV/Batt/Diesel based hybrid system. In this study, a minimum SOC of 20% has been considered. Once the battery SOC reaches its minimum level and load demand is below the minimum starting threshold of diesel generator (25% of the maximum capacity of the generator), the load is then unmet. In the study, in some configuration HOMER software chose small capacity wind turbine (1 kW only) because of its higher capital cost and the velocity availability. The higher capital cost does not allow Case III to be economically feasible over Case I. Halabi et al. (2017) studied a performance analysis of PV/Batt/Diesel based hybrid energy system and found the COE of 0.30$/kWh and the NPC of $9345,510 for the optimized configuration. Mamaghani et al. (2016) analyzed techno-economic feasibility of PV/Wind/Batt/Diesel hybrid energy system for stand-alone rural electrification in Colombia and reported the COE and NPC at Unguia location 0.44$/

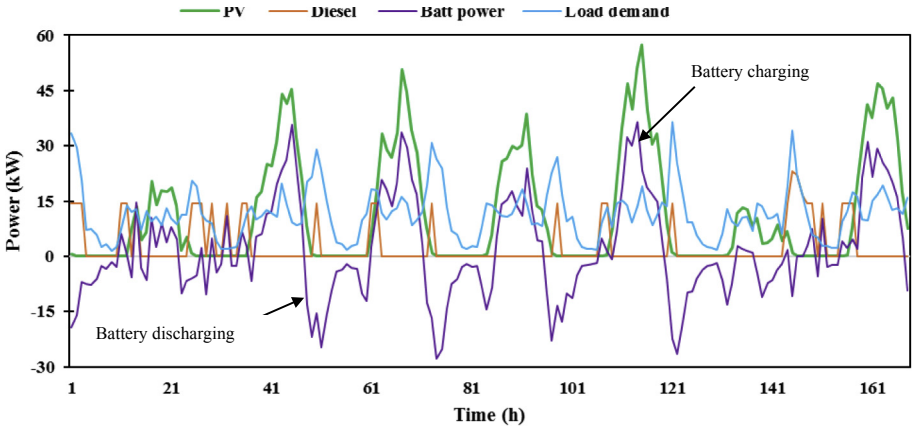


Fig. 11. Load meeting using different systems over a week (July).

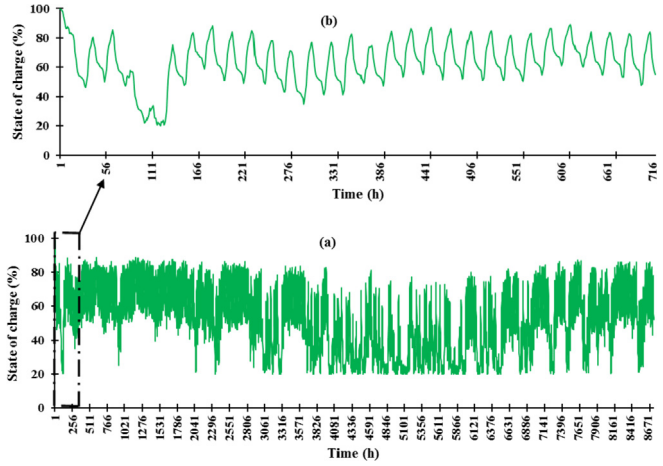


Fig. 12. Battery state of charge (a) over a year, (b) over one month (January).

kWh and $372,736, respectively with the renewable penetration of 98%.

3.8. Sensitivity analysis

The results of this study are further extended to examine the effects of variation of different input parameters on COE and NPC of the optimized (PV/Batt/Diesel) hybrid system. In this context, the effects of varying the discount rate, fuel cost, capital cost, PV, Diesel Genset, and battery bank costs on the COE and NPC are presenting in Fig. 13. The data presented in Fig. 13(a) and (b), 100% (COE = 0.37$/kWh and NPC = $357,284) in horizontal-axis refers to input parameters considered in this study for analysis (baseline scenario). On the other hand, anything less than 100% is the reduction in costs from the baseline scenario, and greater than 100% is the increase in costs from the baseline scenario. It is evident from Fig. 13(a) that the variation of overall capital cost and discount rate may have significant effects on COE. A 20% and 40% reduction in capital cost decreases the COE around 0.31$/kWh (16%) and 0.26$/kWh (30%), respectively in comparison to the base case (0.37$/kWh) whereas 20% and 40% rise would increase the COE almost 0.43$/kWh (16%) and 0.45$/kWh (22%), respectively. Furthermore, a 20% and 40% drop in the discount rate would reduce the COE 0.33$/kWh (11%) and 0.30$/kWh (19%), respectively from the base and 20% and 40% further increase would augment 0.40$/kWh (8%) and 0.43$/kWh (16%), respectively, than the base case scenario. Although the variation of diesel generator set and fuel costs have an insignificant effect on COE, the PV module, and battery cost have marginal effects on the COE. The similar trend is true for the variation of the above parameters on NPC as shown in Fig. 13(b). However, the variation of discount rate has significant effects on the NPC where a 20% decrease in the discount rate would increase around 6% and a further 20% drop would increase NPC around 17% than the base case ($357,284). Additionally, a 20% rise in the discount rate would reduce the NPC by 9%, and a 40% rise would reduce the NPC by 15%. On the other hand, 20% and 40% rise in capital cost would increase the NPC by 15% and 21%, respectively, compared to the base case ($357,284), whereas 20% and 40% reduction would decrease 16% and 31%, respectively from the base case scenario.

3.9. Optimized system (PV/Batt/diesel) vs. grid-connected and solar home system (SHSs)

Apart from the comparison of the different Cases which are

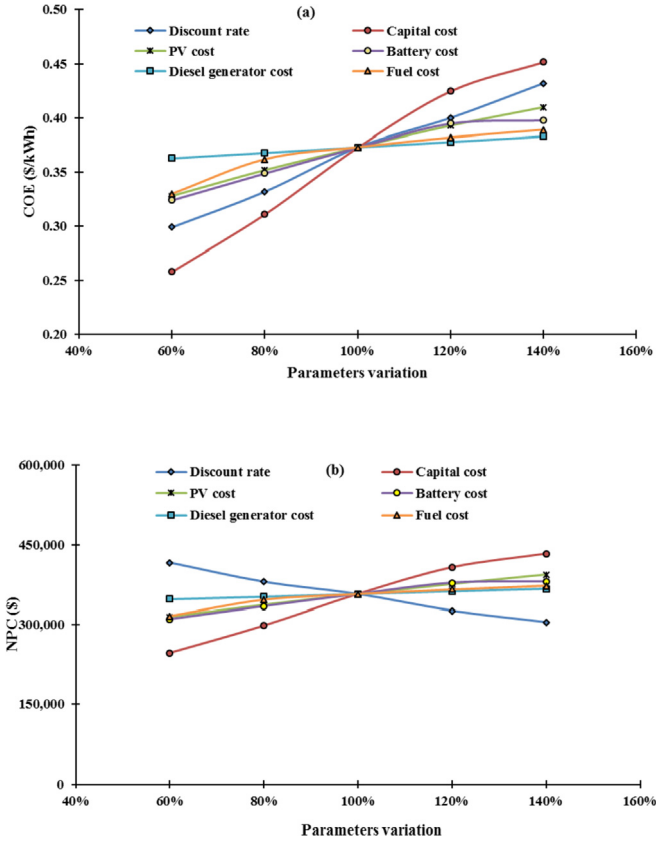


Fig. 13. Sensitivity analysis of (a) Cost of Energy (COE), and (b) Net Present Cost (NPC) for the optimized hybrid system (PV/Batt/Diesel generator).

categorized by the HOMER software tool based on COE and NPC, the study further analyzed the optimized hybrid system with existing national grid and SHSs. This comparative analysis would be helpful for the policymaker to decide on the hybrid based energy system. The COE of selected optimal configuration is 0.37$/kWh which is quite high concerning the cost of national grid connection. The Government purchase electricity from the grid-connected diesel power plant at cost 0.34$/kWh and grid-connected Heavy Fuel Oil (HFO) based plants at cost $0.11–0.22$/kWh (Maleki, 2018). From the above comparison, it is quite clear that this system not possible to cope with the grid cost. However, if the electricity connection cost from the grid to the selected area is considered, then the differences of COE may reduce. On the other hand, the cost of SHSs is approximate 0.72$/kWh (Bhattacharyya, 2015). Additionally, the total cost of installation and maintenance of the solar with battery system (SHSs) is almost double ($573,602) than the hybrid energy system ($308,977) as shown in Table 5. Even the SHS does not provide economic benefits as well as the electricity equivalence to the hybrid energy system. Due to the geographical position, as it is quite ardent to introduce grid connection in the remote area in the near future so the hybrid energy system will be proved beneficial in comparison to single PV system or diesel generator system only. Moreover, regarding cost and energy demand of the hybrid energy system is far better than the SHSs and single diesel generator system. Based on those criteria, there is no other possible efficient option for the remote area rather than selecting the hybrid energy system.

3.10. Economic benefits

The proposed hybridized system (PV/Batt/Diesel) has the potential economic benefit (COE = 0.37$/kWh) as compared to the SHSs (COE = 0.72$/kWh) as discussed above. Additionally, this system has the significantly lower cost of energy than the PV/Batt (0.574$/kWh) and the Batt/Diesel (0.641$/kWh) system. Furthermore, sensitivity analysis shows that a 40% drop in the discount rate would cut 19% COE from the base case scenario; in this regard, government and the support from NGO's would be appreciable. Hence, the benefits of selling the electricity to the local consumers and the national grid would help to reduce the excess energy substantially.

3.11. Environmental benefits

Environmental benefits of the system are assessed based on the number of the pollutants released by the system. Contaminants including CO_2, CO, NO_x, SO_2, and the unburnt hydrocarbon are the main culprits for the environmental pollution released by the system. In the selected area, the consumption of kerosene is the main contributor to the environmental pollution. The CO_2 emission from kerosene is 2.5 kg/l (Hoque and Das, 2013a). The kerosene consumption of the selected area is 0.045 l/h and operating hour is assumed as 4 h. Therefore, total kerosene consumption is calculated as 14,454 l/yr for 220 households, which accounted a total of 36,135 kg/yr of CO_2 as shown in Table 8. Regarding total electricity supplied (88,354 kWh) from the grid to the off-grid area, the rate of CO_2 emitted from the grid electricity is 41,085 kg/yr taking the electricity specific factors of grid electricity of Bangladesh as 0.465kg/kWh (Nandi and Ghosh, 2010a). However, the proposed hybrid system is emitted CO_2 approximately 13,720 kg/year. It is also obvious that CO_2 emission from the optimized system is significantly lower than the diesel-only system. From the above discussion, it is evident that the streamlined system reduces CO_2 emission by around 62% in comparison with the kerosene used in a current situation and 67% with the grid connected system. Therefore, in contrast, the hybrid system will contribute much more in the case of having a green environment.

3.12. Social benefits

Electricity is the basic need in today's world, which can make a revolutionary change in the socio-economic life of a person as well as in the total community. In this study, the social benefits of electrifying the areas are reported based on the field visit to the selected area, consulted with expert opinions, personal judgments, and reviewed published literature. It is observed that electricity is the crucial need for the local people to improve their socio-economic status. Access to electricity infrastructure is a necessary step in modern activities and plays a significant role in boosting the socio-economic conditions of a society (Jimenez, 2017). Additionally, electricity services allow the community members, mainly women and children to the best use of time in household activities and education, thus increasing the household incomes (Lipscomb et al., 2013; Grogan and Sadanand, 2013). Renewable energy technologies have more job opportunities (three times) than the fossil fuels or nuclear-based power generation units (Islam et al., 2008). Rural electrification can be used as an effective anti-deforestation policy as rural community mostly use woods and forest residue for cooking (Tanner and Johnston, 2017). The smoke produces from the lamp and burning biomass for cooking is causing severe damage to health. Moreover, lack of pure drinking water leads the people towards the severe health-related problem. To provide the drinkable water and water pumping system for the better irrigation there are no alternatives to the electrical connection. According to studies, after having electricity users can work even after sunset which can help them to earn extra money from different sorts of activities, can improve their social life and status (Mondal and Klein, 2011; Urmee and Harries, 2011). Electricity also allows area people to be updated with the modern world by introducing radio, television as a source of information and entertainment (Malakar, 2018). In summary, it is expected that hybridized power system can contribute to the rural community with better living standards, raising economic activities, empowering women, safety, employment opportunities, and sustainable development.

3.13. Implementation challenges

Although rural electrification has numerous benefits, the implementation of the hybridized energy system in off-grid areas is not an easy task. There are several challenges such as technical, political, economic, and socio-cultural those need to overcome to make this attempt a successful one.

- The design and implementation of a hybridized power system is a complex one and a relatively challenging task because of the system topology (Mohammed et al., 2015; Nehrir et al., 2011). Additionally, the hybridized system needs higher capital investment costs compared to the diesel only system (Table 5). The market infrastructure to support the hybrid system components is insufficient. Furthermore, the timely maintenance of a large number of batteries and the availability of spare parts in these remote areas also is an issue (Hoque and Das, 2013b).
- Policy and regulatory is one of the major issues in developing countries like Bangladesh for making any project successful. Sustainable and affordable energy access to the paramount of alleviating poverty in developing countries (Terrapon-Pfaff et al., 2014). Renewable energy policy of Bangladesh offers some financial incentives such as 15% VAT exemption from purchasing equipment, and raw materials and 10% higher purchasing price from the private sectors (Renewable energy policy o, 2018). However, it unclear whether the same benefits will apply to the hybrid energy system or not. It is clear that having good energy resources does not make the system a successful one; it is the proper guidelines policy and regulatory towards the installation of the system.
- Implementation of the hybrid system is not possible by going beyond the ongoing rules and regulatory but can be done by reconstructing the new law. Political commitment from government is crucial for the rural electrification with the renewables (Urmee and Md, 2016). However, in the current institutional structure, it needs approval from different ministries, agencies, and institutions. Therefore, without strong coordination with them, it is impossible to get permission for a standard period.
- Economic consideration is one of the significant factors to implement such a system in the rural areas, as it is not practically possible without financial assistance. The primary challenge of the developing countries like ours is to manage the high installation cost for the hybrid system compared to the conventional power generation technologies (Nehrir et al., 2011). Additional investment is required for people to be trained, as the system is relatively complicated. However, this hybrid system would create more job opportunities and helps the area to be dependent (Ellabban et al., 2014).
- The acceptance of a technology which involves complexity in it also depends on the socio-cultural freedom as it requires the level of understanding of such technology (Urmee and Md, 2016). In this regard, members of the society and their cultural

background has a strong influence to establish the hybrid system in the remote area. However in many cases, the local people feel confused to come forward for help because of the superstitions, the cultural barrier to accept the new technology (Egbue and Long, 2012).

In this regard, government and private organizations should come forward to overcome these hurdles to implement the standalone hybridized system. Integration of renewable energy resources to the conventional system not only reduce the greenhouse gas emissions (22% than diesel-only system) but also improve the reliability (than SHSs) of supplying electricity to meet the required load demand.

4. Conclusion

The present study investigates the potential application of the hybridized system for the electrification of a remote community by simulating an effective hybrid energy system with optimum cost evaluation and control emission of harmful products. In this study, six different cases of hybrid systems are studied and an optimal configuration is selected on the basis of the lower Cost of Energy and pollutant emissions and is compared the relative advantages and disadvantages with the grid connected and Solar Home Systems (SHSs). This research also investigates the significant results of different configurations regarding Life Cycle Emissions, Duty Factors, Renewable Penetration, and other economic and environmental parameters. Among the various system configurations, a PV/Batt/Diesel generator–based hybrid system with PV module capacity of 73 kW, a 57 kW diesel generator set, and a 373 kWh battery bank is the most optimally sized configuration while meeting 88,354 kWh annual load demand. Social perspectives and challenges also discussed. The primary outcomes of this study are summarized as follows:

- In this sizing optimization, the Cost of Energy for optimal hybrid system configuration (Case I) is 0.37$/kWh with Net Present Cost of $357,284. Although the COE and the NPC are comparable with the Case II, the higher capital and replacement costs along with environmental pollutants emissions make less favorable than Case I. The proposed system is however costly than national grid connection but has a lower price compared to the Solar Home Systems (SHSs) installed in Bangladesh.
- From the analysis, it is also evident that the optimized system has lower operational CO_2 emissions (13,720 kg/yr) than kerosene-based lighting (36,135 kg/yr) and the grid-connected electricity (41,085 kg/yr). Although Case III and Case IV have zero operational emissions and lower Life Cycle Emissions than Case I and Case II, these systems have much higher cost than Case I and Case II. The optimized system (Case I) has lower Duty Factor (DF) compared to the Case V and Case V.
- In relation to Renewable Penetration (RP), Case III and Case IV have higher RP (100%) among the system configurations. On the other hand, Case I and Case II have comparable renewable penetration (89%) whereas Case V and Case VI have almost zero renewable penetration. It is attributed to higher operational as well as the Life Cycle Emissions.
- Sensitivity analysis shows that the COE is highly sensitive to capital cost and discount rate. However, the variation of fuel price, PV module cost, battery cost, and diesel generator cost have less effect on the COE. A similar trend is also true for NPC but inversely for variation of the discount rate.

While the optimal system has higher COE than the grid-connected system, and the grid connection is far away from reality for a decade, the benefits of electrification in the remote areas like this not only improve the quality of life but also contribute to the overall economic development of the country. The outcome of this research could be a step forward to look into the hybrid energy technology for rural electrification in the northern area of Bangladesh. Although the study focuses on the detailed technical, economic, environmental, and social perspectives of a hybridized energy system, the social assessments regarding the ability of the rural community to pay the electricity cost is worthy of future investigation.

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Nomenclature

A_w: Wind turbine swept area (m^2)
B_{soc}: State of battery charge (%)
c: Battery capacity ratio
C_A: Annualized cost of system components ($)
C_{A_cap}: Annualized capital cost ($)
C_{A_rep}: Annualized replacement cost ($)
$C_{A_O\&M}$: Annualized Operation and Maintenance cost ($)
C_B: Charge capacity of the battery bank (Wh)
C_{Batt}: Sum of the present value of the capital and maintenance costs of batteries ($)
C_{NPC}: Total net present cost ($)
E_S: Total energy generation (kWh)
E_L: Energy generation or stored in each component (kWh)
F: Fuel consumption rate (kg/h)
F_0: Fuel curve intercept coefficient ($m^3/h/kW_{rated}$)
F_1: Fuel curve slope ($l/h/kW_{output}$)
f: Annual inflation rate (%)
f_{PV}: De-rating factor (%)
H: Hub Height (m)
H_{ref}: Reference Height (m)
I_{bat}: Battery Current (A)
I_T: Solar radiation incident on the PV array (kW/m^2)
I_S: Incident solar radiation at standard test condition (1 kW/m^2)
$I_{T,NOCT}$: Solar irradiation (kW/m^2) at NOCT
i: Real annual interest rate (%)
i': Nominal interest rate (%)
k: Battery rate constant
N: Project lifetime in years
$N_{s/s}$ Number of start-stop
P: Power ration of the battery bank
P_{dg}: Electrical output of the generator (kW)
P_{in}: DC input power to the inverter (kW)
P_{out}: AC output power (kW)
P_r: Rated power of wind turbine (kW)
P_{wt}: Output power of a wind turbine (kW)
Q: Total amount of battery energy at the beginning of time step (kWh)
$Q_{bat,0}$: Initial battery charge (kWh)
Q_1: Available energy in the battery at the beginning of time step (kWh)
Q_{max}: Total capacity of the battery bank (kWh)
Δt: Length of the time step (h)
T_a: Ambient temperature (°C)
T_c: PV cell temperature (°C)
T_s: PV cell temperature under standard test condition (25 °C)
$T_{a,NOCT}$: Atmospheric temperature at NOCT (°C)
$T_{C,NOCT}$: Nominal operating cell temperature (°C)
U_L: Coefficient of heat transfer to the surroundings (kW/m^2 °C)
V: Wind velocity at height H (m/s)
V_2: Cut-out velocity (m/s)
V_1: Cut-in velocity (m/s)
V_r: Rated wind velocity (m/s)
V_{bat}: Battery Voltage (V)
V_{ref}: Wind velocity measured at the reference height H_{ref} (m/s)
Y_{dg}: Rated capacity of the generator (kW)
Y_{PV}: Rated Capacity of PV Array (kW)

Greek symbols

τ: Solar transmittance (%)
α: Solar absorptance (%)
$\tau\alpha$: Effective transmittance-absorptance of PV array
α_P: Temperature coefficient of power
β_i: Lifetime equivalent CO_2 emission of each component (kg CO_2-eq/kWh)
γ: Ground surface friction coefficient
η_{inv}: Efficiency of an inverter (%)
η_{PV}: PV panel efficiency (%)
η_w: Wind turbine efficiency (%)

Abbreviation

AC: Alternating Current (A)
COE: Cost of Energy ($/kWh)
CRF: Capital recovery factor
DC: Direct Current (A)
DF: Duty Factor (kWh/start-stop/yr)
EE: Excess Energy (kWh)
GHG: Green House Gases
HOMER: Hybrid Optimization Model for Electric Renewables
LCE: Life Cycle Emission (kg CO_2-eq/yr)
NASA: National Aeronautics and Space Administration
NOCT: Nominal Operating Cell Temperature (°C)
NPC: Net Present Cost ($)
PV: Photovoltaics
RP: Renewable Penetration (%)
SOC: State of Charge (%)